

ANATOMY OF A CHIP

Semiconductor manufacturing involves multiple specialised stages, but chip design anchors the entire ecosystem

STEP 1 >> CHIP DESIGN

Shaping product requirements into a chip design that sets what the chip does, how fast it runs, and how efficiently it uses power.

Global leaders: Qualcomm, NVIDIA, AMD, Broadcom, MediaTek

STEP 2 >> PRE-SILICON VALIDATION

Ensuring the design works as intended by digitally testing logic, performance, and power before manufacturing begins.

Global leaders: Synopsys, Cadence, Siemens EDA

STEP 3 >> WAFER FABRICATION

Physically manufacturing the chip exactly as designed, using complex processes on silicon wafers such as advanced lithography and deposition processes.

Global leaders: TSMC, Samsung, Intel, GlobalFoundries

STEP 4 >> WAFER-LEVEL BUMPING & TESTING

Preparing manufactured wafers for advanced packaging and verifying electrical performance early.

Global leaders: TSMC, ASE Technology, Amkor

STEP 5 >> ASSEMBLY, PACKAGING & TESTING (ATMP / OSAT)

Turning fabricated wafers into usable chips by packaging and testing them to design specifications.

Global leaders: ASE Technology, Amkor, JCET